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How We Found It, and How We'll Build It

The full research trail + the technical architecture of the AI Voice Front Desk — for internal alignment.

Audience: internal (founder + Shrikaanth)

Prepared: June 2026

Companion files: Data Bible (21 tabs) · Build Bible

Scope: research method & findings · system architecture · self-improvement · the one-operator model

Executive Summary

This is an internal memo. It documents two things in depth: **(1) exactly how we researched the market** — every dataset, keyword, search volume, trend, SEO score, subscription price, and customer complaint we reviewed — and **(2) exactly how the product works technically** — the architecture, the self-improving feedback loop, and how one person plus AI agents run it. Finance is intentionally compact.

560

KEYWORDS RESEARCHED

164

COMPETITORS TEARDOWN'D

4 yrs

OF DEMAND DATA

The research pointed, with evidence, to a single opportunity: an **AI voice front desk** for appointment-driven local businesses. The rest of this memo shows the work behind that conclusion and the engineering plan to deliver it.

PART I

The Research

How we found the spot — every step, dataset, and number.

Method & Scope

We ran a structured demand × supply analysis, deliberately cross-validated, before committing to any idea.

#	What we did	Source / tool	Volume
1	Pulled 4 years of search demand (volume, trend, seasonality, CPC, competition)	Google Keyword Planner	3 geographies
2	Pulled SEO difficulty for every keyword (winnability)	Ubersuggest (Neil Patel)	560 keywords
3	Built a competitor teardown (valuation, pricing, GTM, architecture, complaints)	Two independent AI engines	164 tools
4	Computed everything into an auditable workbook	The Data Bible	21 tabs

Why two AI engines for the teardown? We ran the competitor analysis twice through different engines and only treated findings as high-confidence where *both independently agreed*. This filters hallucination and bias — the most-corroborated finding (poor data quality + punishing pricing) became our strongest signal.

Method caveats (kept honest): Planner volumes are bucketed → read as ranges; market sizes are planning estimates; competitor specifics are AI-generated and cross-validated, not audited filings.

The Keyword Universe

We did not eyeball a few terms — we built and scored a structured set of 560 keywords across three markets.

Geography	Keywords	Total avg searches / month
Worldwide	559	25,135,650
United States	517	6,013,630
Canada	302	910,840

Each keyword carries: average monthly searches, three-month change, **year-over-year change**, **the full 4-year monthly series** (so we computed seasonality and per-year trend), competition, top-of-page CPC (low/high), and — added later — Ubersuggest **SEO difficulty** and paid difficulty.

The buckets we searched

The keyword set was organised into 9 intent buckets so we could separate noise from real demand: category head terms, problem/JTBD terms, ICP-qualified terms (SMB / mid / agency), agency-specific, all-in-one/consolidation, comparison/switching, client-delivery, and bare one-word head terms (crm, invoicing, scheduling...).

Critical methodological catch: our first dataset looked huge but was **~84% branded search** — people typing the names of products that already won (notion, quickbooks, calendly). That measures fame, not unmet demand. We re-pulled the research around *non-branded, problem-based* terms — which is what surfaced the real opportunity.

What the Demand Data Showed

Real category demand (non-brand), with winnability

Keyword	US searches/mo	SEO difficulty	CPC (high)
knowledge base software	301,000	59	\$81
crm software	90,500	70	\$73
help desk software	18,100	48	\$122
business management	27,100	29	\$48
client management software	6,600	34	\$77
cloud based business phone system	6,600	9	\$76

4-year momentum — where attention is moving

Rising	4-yr trend	Declining	4-yr trend
brevo	+6,320%	convertkit	−70%
gohighlevel	+573%	freshdesk	−45%
apollo.io	+346%	mailchimp	−33%
close	+116%	zendesk	−24%

Seasonality & commercial intent

The 48-month series let us compute seasonality per keyword (e.g., finance/accounting terms peak in Q1 tax season; most SaaS demand is steady year-round — good for subscriptions). CPC told us where money is: the most expensive clicks clustered around **all-in-one/agency** (gohighlevel CPC \$1,415 — **highest in the dataset**) and **support/help-desk** (\$80–126).

The grey space: real category demand at **SEO difficulty 8–35 with \$60–130 CPCs** — high commercial value, almost nobody ranking. Meanwhile the giant head terms (crm 90, notion 77, forms 93) are unwinnable walls.

The Competitor Teardown (164 tools)

For the supply side we tore down 164 tools across 14 categories — valuation, pricing, architecture, and, crucially, their customers' biggest complaints.

~70%

INCUMBENTS ARE LEGACY

16 / 16

BOTH ENGINES FLAG BAD DATA

24 / 79

COMPLAIN 'MISSING FEATURES'

Subscription pricing we recorded (sample)

Tool	Price	Tool	Price
Begin by Zoho	\$7–12 /user/mo	Apollo.io	\$49–119 /user/mo
Pipedrive	\$14–99 /user/mo	Close	\$49–139 /user/mo
Folk	\$18–39 /user/mo	Cognism	\$3,000–6,000+ /user/yr
Streak	\$15–129 /user/mo	Capsule	\$18–54 /user/mo

Customer feedback / complaints (verbatim, coded)

Tool	Biggest complaint
Lusha	"Credits disappear fast on wrong or outdated rows"
Seamless.AI	"High volume of unverified or generic catch-all emails"
Close	"High telephony usage costs outside base monthly plan"
Folk	"Lacks advanced pipeline automation and conditional routing"
Streak	"Browser extension reliance causes instability"

Valuations recorded too (e.g., Pipedrive \$1.5B, Apollo.io \$1.6B) — confirming these are well-funded incumbents we must out-flank, not out-spend.

How the Spot Was Found

We overlaid the four lenses on every keyword and tool, and the conclusion fell out of the data.

Lens	Source	What it told us
Demand	Keyword Planner	How many people want it + trend + seasonality
Winnability	Ubersuggest SEO	Whether we can realistically rank/win it
Commercial value	CPC	How much the market pays to acquire here
Incumbent weakness	Teardown complaints	Whether the existing tools are beatable

The Opportunity Score (top in-lane results)

Keyword	US vol	SEO	CPC	Score
social media reporting tool	720	10	\$106	80.8
client retention software	880	15	\$126	79.5
project management software for agencies	480	8	\$72	76.4
crm with client portal	210	17	\$105	75.7
business management	27,100	29	\$48	74.1
client management software	6,600	34	\$77	73.8

All four lenses converged on the same territory — the **live, voice-first customer-intake layer**: highest CPCs, the single most open keyword (phone system, SEO 9), declining incumbents, and a "why now" (real-time voice AI matured in 2025). That is the AI Voice Front Desk.

PART II

The Execution

How the system works technically, how it self-improves, and how one operator runs it.

The Product — One Engine, Two Modes

Model 1 — Inbound Receptionist

Answers calls to the business 24/7, answers from the business's knowledge base, qualifies, and books into their calendar with SMS confirmation and human handoff when unsure.

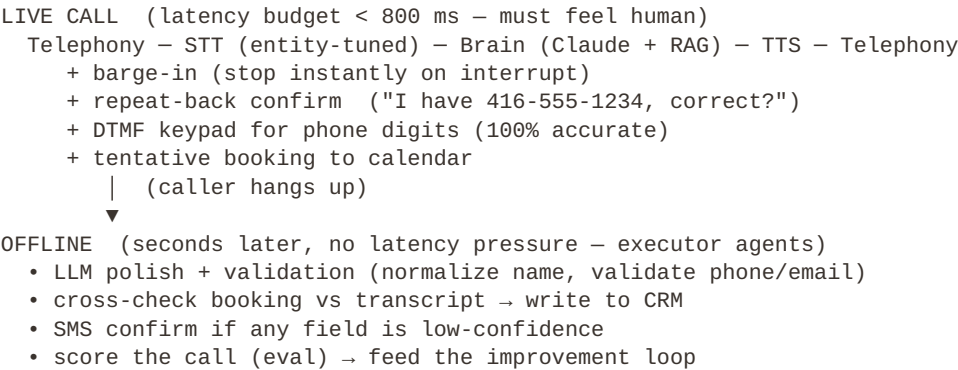
Model 2 — Speed-to-Lead Caller

When a lead submits a paid-ad form (with consent), the AI calls back within 60 seconds, qualifies, and books — capturing the ~390% conversion advantage of the 5-minute rule.

Both modes are the **same engine** (brain + knowledge base + booking adapter). Model 2 = Model 1 + a lead-webhook trigger. Build once; point it at inbound calls, lead callbacks, or both.

The Technical Architecture

A real-time voice pipeline with a deliberate split: the live call stays lean; the heavy intelligence runs offline.



The modules (each swappable behind an abstraction layer)

Module	Job	Tool (Phase 1)
Telephony	Carry the call	Twilio / Telnyx
STT (ears)	Speech → text, capture entities + accent	AssemblyAI / Deepgram
Brain	Reason, qualify, decide (reads knowledge base)	Claude (Haiku → Sonnet)
TTS (mouth)	Text → natural voice, stop on interrupt	Cartesia / ElevenLabs
Orchestration	Wire it all in real time, barge-in	Retell → LiveKit (Phase 2)
Booking + SMS	Write to calendar, confirm	Google Cal / Jane / Jobber + Twilio
Capture + Eval	Log every call, score it	Supabase + Coval

Phase 1 ships fast on Retell (compliance + integrations included); Phase 2 migrates the core to LiveKit/Pipecat to own interruption handling, a multi-vendor router, and a fine-tuned accent model — without a rewrite, thanks to the abstraction layer.

How It Self-Improves (Auto-Correct)

The model gets smarter every night — through a closed, eval-gated loop, with a human approving changes.

1. CAPTURE every call → transcript + audio + outcome + confidences
▼
2. AUTO-EVAL LLM-as-judge scores each call: booked? mis-heard a number? interrupted rudely? understood the accent?
▼
3. MINE cluster the failures into patterns
("missed suburb names", "didn't stop on interrupt"...)
▼
4. PROPOSE FAST (no training): add missing FAQ to knowledge base, add mis-heard words to the STT boost list, add a winning call as an example, tune the barge-in threshold.
SLOW (periodic): fine-tune an accent model on our own accented calls; distil our best calls into the brain.
▼
5. GATE replay the change against a saved test-set of real calls.
Ship ONLY if it improves the score and breaks nothing.
▼
6. APPROVE Shrikaanth approves → live. Loop repeats.

Why this is also the moat

The loop runs **per vertical**. Every physio call (anonymized) improves the "Physio Brain"; every construction call improves the "Construction Brain." Each new client makes the product smarter for the next — a **data network effect** a fresh competitor cannot replicate, and a generic foundation model will never build per-vertical.

Honest boundary: fully autonomous self-editing is unsafe today, so the design is *auto-detect* → *auto-propose* → *eval-gate* → *human-approve*. The eval gate is the seatbelt that makes auto-improvement safe. As trust grows, more changes ship automatically.

How One Person Runs Everything

The company is operated by one human (Shrikaanth) leveraging a team of AI executor agents. The human reviews and approves; the agents do the repetitive execution.

Executor Agent	What it does	Built/Rented
Dev Agent	Writes the app, integrations, webhooks	Claude Code
Onboarding Agent	Fetches client's domain + Google profile + docs → drafts config → client confirms (<1 day)	Build
QA / Eval Agent	Scores every call, flags failures	Coval + Claude
Improvement Agent	Proposes knowledge/vocab/prompt fixes	Build
Follow-up Agent	Drafts post-call SMS/email	Build
Sales-Research Agent	Researches prospects, drafts outreach	Build
Monitoring Agent	Watches errors/outages, alerts	Rent + wire

The human's role = reviewer/approver, not doer. Onboarding auto-drafts; QA flags only the calls that need eyes (not all of them); improvements are proposed and wait for one approval. We hire our first additional human only at ~30–40 customers.

This is the structural bet: **AI agents are the workforce; the human is the orchestrator.** It is what makes a billion-dollar-scale ambition feasible for a team of one to start.

Our USP

Five things, together, that no single competitor combines:

1. **Vertical-deep, not generic.** A "Physio Brain" / "Construction Brain" that knows the industry's language, scheduling rules, and integrations — vs. a one-size-fits-all bot.
2. **Actually understands the caller.** Accent-adapted speech, fine-tuned on our own call data — built for our multicultural / realtor markets where generic American-English bots fail.
3. **It books, it doesn't just talk.** Belt-and-suspenders data capture (repeat-back + keypad + validation) so names, numbers, and emails are right — the whole point of a booking.
4. **Self-improving + compliant by design.** Gets smarter every night; HIPAA/consent/anti-bias built in from day one.
5. **Run by one + agents.** A lean, AI-leveraged operation = low cost, fast iteration, healthy margins.

"The AI front desk that actually understands your callers, books them correctly, gets smarter every night — and is run by a team of one plus its own AI agents."

The Numbers, Briefly

Item	Figure
Pricing	\$199 / \$399 / \$799 per location/mo + \$300–500 setup; optional \$15–40 per booked appointment
Delivery cost	~\$0.10–0.18 / minute → ~\$75–225 / location / mo at typical volume
Gross margin	~70–80%
Run all 7 of our own businesses	~\$500–1,600 / mo (usage)
Cost to first external paying customer	~\$5,000–15,000 (incl. one-time legal)
North Star metric	Qualified appointments booked / month

Detail and pricing models live in the Build Bible; this memo's focus is research + execution, so economics are summarised.

Where We Stand & Next Step

- **Research:** complete — 560 keywords scored across 3 geographies, 164 competitors torn down, all computed in the 21-tab Data Bible.
- **Conclusion:** locked — AI Voice Front Desk; physio first (our clinics), then construction, then realtor/mortgage; PPC Guru as pilot #0.
- **Technical plan:** architected — Retell now, LiveKit later; real-time/offline split; self-improving loop; one-operator + agent org.
- **Compliance:** mapped (US + Canada), handled by design.

Immediate next step — Phase 0: stand up the stack + abstraction layer, build the call-capture/eval schema, sign vendor compliance agreements, and draft consent/disclosure. Then the PPC Guru speed-to-lead pilot to prove conversion lift on our own leads.

Appendix — source files (committed, auditable)

Data Bible (21 tabs) · Demand keyword + Ubersuggest set (560 kw) · Build Bible (architecture, tool matrix, regulatory register, agent org) · Strategy Findings. Every figure in this memo traces to these.

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